

On Chain Driven, Adaptive, Underactuated Fingers for the Development of Affordable, Robust Humanlike Prosthetic Hands

Trevor Heinmann, Raymond Wallace, and Minas Liarokapis

Abstract—Amputations and limb loss can have detrimental effects on personal well-being. Although prosthetic devices can offer significant benefits helping amputees regain some of the lost dexterity, they often lack the required affordability and durability. Current affordable prosthetic designs have trended towards underactuation which leads to stable grasps but is often characterized by low durability. In this paper, a new chain-driven, adaptive, underactuated finger design has been proposed for the development of affordable and highly durable prosthetic hands. The transmission mechanism employed is in the form of a steel roller chain routed through sprockets. The finger phalanges are constructed of 3D printed PLA, and finger flexion is produced by pulling the internally routed roller chain. In total, six 3D printed PLA sprockets are used for chain routing, with an emphasis on high force transmission. The performance of the proposed chain-driven finger was experimentally validated and compared with an analogous tendon-driven version. The metrics employed for this comparison were longevity, pinch grasp efficiency, force response, and maximum force capability. The chain-driven finger was shown to have a higher maximum transmissible force, better long term durability, and no issues related to elongation (such as tendon elongation). The cost to manufacture the chain-driven robotic finger is only 91 USD, making it an excellent solution for affordable prostheses.

I. INTRODUCTION

The human hand is largely regarded as Nature's most versatile end effector, able to complete a staggering number and variety of different tasks in our everyday life with an astounding effectiveness. Unfortunately, the extraordinary utility of the human hand is understood most keenly by those suffering from upper limb loss. Upper limb loss can be a traumatic experience and an extreme detriment to physical and mental wellbeing [1]. It is a growing concern, with over 500,000 cases of upper limb loss reported in 2005, a number is expected to double by 2050 [2]. This demonstrates a clear and growing need for prosthetic devices, which can restore some amount of lost functionality [3], [4] and increase the confidence and autonomy of the user [5]. Some often overlooked aspects of prosthetic hands are the durability and longevity with which they perform tasks. Barring injury or medical condition, the human hand will remain operational for an entire human lifespan, a marked difference from most currently available prostheses which generally last 3 years in the USA [6], or an estimated 300,000-1,000,000 open and close cycles depending on usage [7]. Another practical consideration in prosthesis design is the cost of the device,

Trevor Heinmann, Raymond Wallace, and Minas Liarokapis are with the New Dexterity research group, Department of Mechanical Engineering, Faculty of Engineering, The University of Auckland, New Zealand. E-mails: {thei442,rwal148}@auckland.ac.nz, minas.liarokapis@auckland.ac.nz

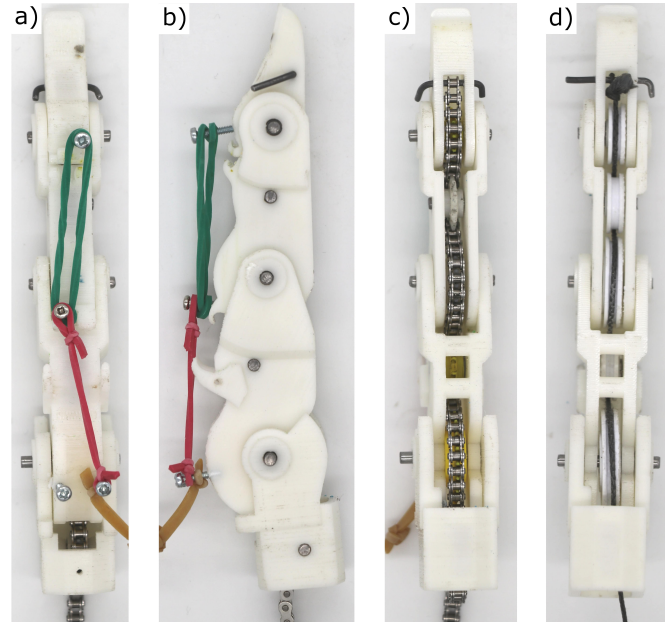


Fig. 1. The proposed chain-driven, adaptive, robot finger. Subfigure a) shows the top view, b) shows a side view, c) shows the bottom view, and d) shows the bottom view of the tendon-driven version used for comparison purposes.

particularly for low-income or developing countries, where cost is a significant barrier to purchasing a prosthesis for those suffering from limb loss [8]. This creates a clear need for prosthetic devices which are both affordable and robust. Current affordable prosthetic devices generally follow the paradigm of using underactuated, compliant designs which allow for secure, adaptive grasps and can be highly compact and humanlike, such as [9] and [10], but often this choice sacrifices durability and longevity. Force transmission mechanisms for realizing these characteristics vary, but the most popular approach is a tendon-driven actuation routed through channels, as in [10], which simplifies the device but eventually leads to tendon deterioration at the points of contact [11].

In this paper, we present a prosthetic finger with a novel force transmission mechanism utilizing an internally routed steel roller chain. This design results in increased maximum force transmission and long term durability of the device, while maintaining the beneficial characteristics of underactuated, adaptive grasping and affordability. The chain is used to initiate finger flexion under tension by producing torque at each finger joint without directly constraining the position of that joint, similar to a tendon or cable.

The proposed finger was designed to be robust enough to remain structurally sound under significant load. Sprocket radius was maximized to take advantage of high maximum force transmission, maximizing output force. The construction of the chain also has several advantages. First, the static and segmented nature of a steel chain allows for precise length and termination of the chain, which will not suffer from drift or elongation over time. Steel roller chains are also manufactured to have much higher strength and longer lifespans than a prosthetic device is likely to require [12]. Roller chains themselves are around 95-98% efficient [13] and when routed over bearing mounted sprockets, achieve mechanical efficiencies of around 85-90% [14]. Additionally, the increased surface area allows loads to be distributed over a larger internal contact area compared to a tendon or cable, reducing force concentration. However, roller chains as prosthetic force transmitters do require a key tradeoff in size, as the internal space requirements of a chain and sprockets result in a less compact overall package.

The rest of the paper is organized as follows: Section II presents a review of related work, Section III details the design of the proposed chain-driven adaptive, underactuated robotic finger, Section IV presents the experimental results, and Section V concludes the work.

II. RELATED WORK

With respect to the affordability and durability of prosthetic finger designs, current and prior work can be categorised by actuation method and force transmission mechanism. For underactuated fingers, either tendon, linkage, or Shape Memory Alloy (SMA) transmission designs have been explored. Fully actuated designs utilize similar transmission mechanisms, but with increased complexity due to the number of actuators required to fully constrain the degrees of freedom present in the device, as in [15]. Each design approach and transmission mechanism necessitates tradeoffs with regards to durability, affordability, dexterity, degree of similarity to a human finger, and maximum grasp strength.

Tendon actuated finger designs such as [9], [10], [16] and [17] utilize compliance and underactuation to provide adaptive grasping capabilities while remaining lightweight and affordable [16]. Common designs incorporate 3D printed polymer structural components and low cost tendon materials, such as Kevlar line. The flexibility of tendons allow for compatibility with both pinned and compliant joint types such as in [17] and [18], unlike linkage mechanisms. Designs with increased complexity and durability can utilize multiple routed tendons, and can increase performance through additional elements such as internal bearings or additional compliance through finger pads, such as in [19]. Tendon designs also have several inherent drawbacks. The tensile failure force of the tendon can be the limiting factor for the ultimate transmittable force through the finger. A typical tendon material such as 1.1mm diameter braided Kevlar line [20] is rated for a theoretical maximum tensile force of 893 N. With an output to input ratio of the finger mechanism of 0.3 or less, this can limit theoretical maximum fingertip output

force to 270 N or less. Additional factors such as termination knots, irregularities in routing contact surfaces, and frictional wear of the tendon fibres can result in stress concentrations or weakened tendons, causing observed tensile failure force to be significantly lower than theoretical tendon rating, as later shown. Tendon-driven designs actuate the finger by generating a torque force acting at the pivot joints between phalanges. The distance between the tendon route and pivot point is directly proportional to the torque exerted at each joint, and subsequently the grasp force of the prosthesis. One disadvantage of the tendon design is low grasp strengths when tendons are routed close to the pivoting point of finger joints such as in [17], [18], [21], [9], [10] and [16]. Typical grasp force of a hand can be <30 N [21] and <10 N for a finger in a pinch grasp. Another disadvantage of tendon designs is tendon wear and fatigue resulting from the contact of the tendon with pulleys or tracks. This phenomenon has been studied in pulley-rope systems [11]. Deterioration of the tendon material eventually leads to fatigue and failure of the tendon, limiting longevity of the prosthesis. Tendons also elongate gradually over time under repeated load, requiring tuning maintenance of the prosthesis through mechanisms capable of ratcheting tendons such as [22]. The elongation of the tendon also occurs temporarily during loading from tendon material characteristics [23], reducing the accuracy in control over finger positioning during flexion. A further drawback of tendon designs is the possibility of tendons escaping pulleys or tracks when no tension exists, leading to system failure.

Examples of underactuated linkage driven finger designs with 3D printed polymer or metal prototypes are [24], [25], [26], [27], [28]. The main advantage of the linkage design approach is underactuated, adaptive grasping in conjunction with mechanically constrained trajectories. Some designs like [24] utilise extension actuation by spring or elastic component whilst others like [25] can be directly actuated to extend the finger by reversal of input actuation. This ability can provide more control for in-hand manipulation [29]. Another advantage of the linkage design is transmission through solid components such as polymer or metal bars and pivots. Therefore, issues with control and fatigue associated with tendons are avoided. The goal of appearing humanlike provides an additional constraint to linkage transmission designs, as the exterior form limits the internal components' size and movement path. This is particularly relevant for linkages as the freedom of size and trajectory of internal components translates to increased performance, leading to a tradeoff between compactness and performance characteristics such as maximum force transmission. This is one disadvantage for linkage designs, as they generally require more internal space and complexity than other transmission methods. Humanlike linkage designs have been shown to transmit forces without failing of up to 35 N [25], [28]. Durability of linkage based fingers over many cycles of flexion and extension have not been well tested, however standard mechanical component wear of transmission components can be expected, such as polymer fatigue [30].

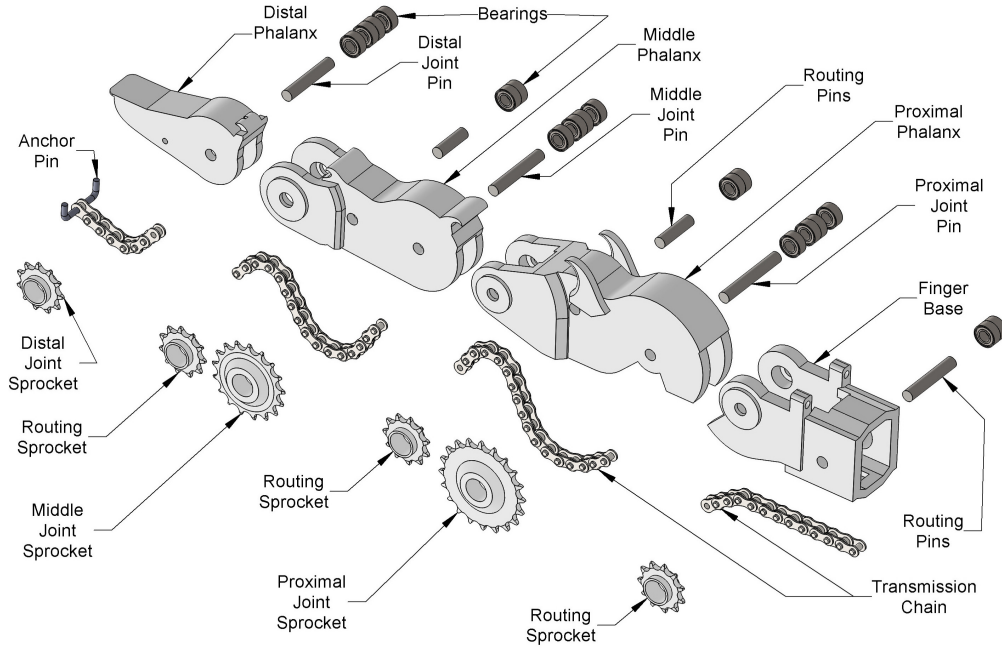


Fig. 2. Exploded view of finger assembly. Each sprocket is specifically sized, with a proximal, middle, and distal joint sprocket of descending size to provide differing torques. The intraphalangeal routing sprockets size is minimized to preserve space. Each sprocket rests on two bearings, and each phalange captures and rests on two bearings at its distal joint. The chain is terminated distally by means of an anchor pin.

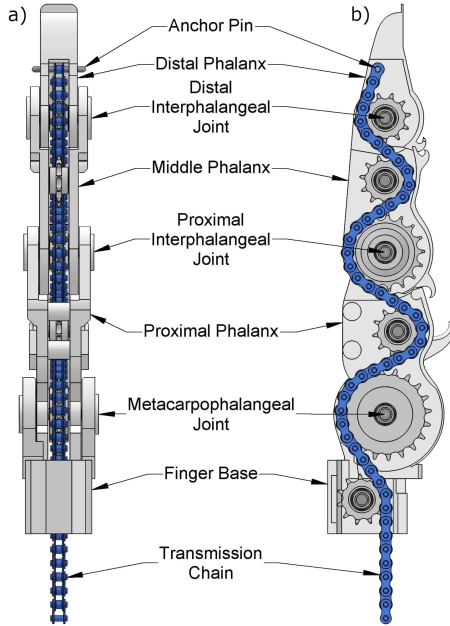


Fig. 3. Chain routing through the finger, with the chain highlighted blue. Subfigure a) shows the chain exposure from the bottom view and subfigure b) shows internal chain routing path and sprockets.

Enhancing durability and maximum transmittable force in a humanlike linkage design finger is possible with higher strength and more durable materials such as metals, as seen in some designs [24], [26]. These materials can increase weight and manufacturing cost, reducing affordability.

SMA's, such as Nitinol wire, can also be used to actuate prosthetic fingers as in [31] and [32]. The benefits of these designs are that they are lightweight and that they provide relatively high actuation forces, as a single strand of Nitinol wire weighing 0.35 g can provide 1 N of force [32]. Disadvantages of SMA designs include difficulty with accurate position control, slow response time, potential robustness issues, and cost of materials. Fully actuated finger designs such as [15] include actuators at each finger joint. The benefits of this design are minimal transmission between actuator and joint, with minimal power loss, allowing up to 30 N force at the fingertip. Negatives of the design include more complicated control requirements and increased weight and cost associated with increasing actuator numbers.

III. DESIGN

A. Overview

The general form and function of the finger is humanlike, with three phalanges and a base section connected via pin joints. Finger flexion is actuated by the application of tension to the internally routed chain. An exploded view of the finger assembly can be seen in Fig. 2. Routing is achieved with six internal sprockets as shown in Fig. 3. Finger extension is passive, actuated via externally routed elastics. Free spinning sprockets proved a very effective solution for reducing chain routing friction. The design intent of the finger was to utilize a roller chain force transmission system so as to produce a prosthetic finger which is humanlike and capable of adaptive grasping while minimizing cost and maximizing robustness. Robustness was defined as long term lifespan and immediate maximum force transmission capability.

This imposed several key design constraints:

- 1) Overall exterior finger dimensions and individual phalanx dimensions are limited by the preference to resemble a human finger
- 2) Dimensions of the roller chain are limited by the smallest steel chains readily commercially available
- 3) Sprocket pitch diameter and tooth width is restricted by compatibility with chain and bearings
- 4) Steel axles must be sufficiently thick to support pin joints and sprockets and PLA structural components and must meet minimum shear and compression strength requirements to account for high chain tension

Internal routing of the chain posed the largest technical obstacle, and drove much of the final design. There were several conflicting requirements; it was necessary to create a routing scheme that encloses the chain as much as possible to prevent debris or interference, it was necessary to prevent contact of the chain with any static components to prevent wear and undue friction, and it was desired to minimize exterior dimensions. This resulted in minimizing internal routing sprockets, other than joint sprockets, and required phalanges larger than human ones. The chain used has a nominal number of 11, with pitch of 3.75 mm, a maximum height of 3.5 mm, and an outer width of 5.44 mm. Additionally, the chain has only one degree of flexibility, necessitating precise alignment between all contact points.

B. Force Transmission

Ball bearings were used at all points of rotational contact to reduce friction as much as possible. The sprockets incorporate two press fit bearings each to allow for free rotation around the pins, as does each phalanx. The sprockets used for testing were manufactured by PLA 3D printing, with more durable SLS nylon versions tested and found unnecessary. It was found that the free spinning nature of the sprockets prevented any transverse forces on the teeth, resulting in only compressive forces being transferred through the sprockets. This allowed for the relatively weak PLA sprockets to be used, rather than aluminium or steel, to reduce cost and allow ease of customisation. The overall profile and outer nominal dimensions of the sprockets were calculated from the pitch and roller diameter of the chain and the internal bore was set to the outer dimensions of the ball bearing. As with all chain sprockets, only certain discrete sizes of pitch diameter were available as a result of needing a whole number of appropriately sized teeth. The designed sprockets are all 1.5 mm thick at the tooth base. The design utilizes four 11 tooth, 13.06 mm pitch circle diameter sprockets for intraphalangeal routing and for the distal interphalangeal joint. The proximal interphalangeal joint pin supports a 17 tooth, 20.40 mm pitch circle diameter sprocket, and the metacarpophalangeal joint pin supports the largest 20 tooth, 23.98 mm pitch circle diameter sprocket. Varying sizes of sprockets were utilized to generate increasing amounts of torque towards the more proximal joints, resulting in sequential joint closing from proximal to distal under uniform passive extension elastics. This allows the finger to more

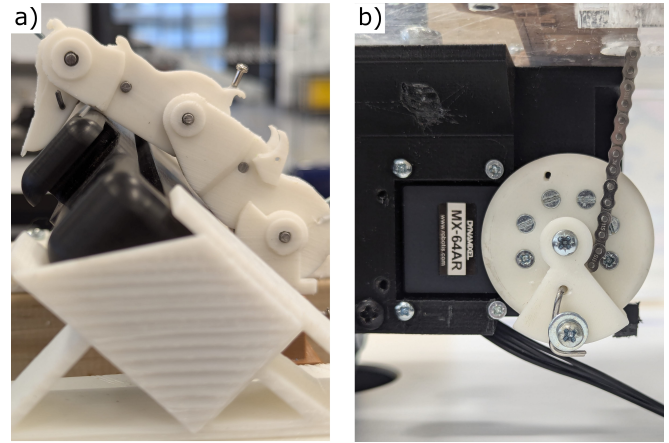


Fig. 4. Test rig adjustments for pinch force testing. Subfig. a) shows the placement of a finger without elastic elements on the force transducer and subfig. b) shows the motor and sprocket mounted to the bottom.

readily exhibit adaptive grasping and conform to the shape of a grasped object. The sprocket tooth profiles are optimized for lack of friction during tooth engagement. This naturally results in thinner teeth which could compromise strength, but in our design the teeth bear no transverse loads, save for in the case of misalignment. The tooth tip profile (when viewed from a plane tangential to the sprocket axis) cutaway angle was made steeper, extending down to the base of the tooth. Improved sprocket tooth shape therefore facilitated self alignment of the chain on the sprocket, providing a margin for error in small inconsistencies in chain alignment.

Exerted grasp force relative to input chain tension was maximized by maximising the size of joint sprockets, particularly the sprocket concentric with the metacarpophalangeal joint. Joint sprocket radius controls the chain routing around the joint, so increased sprocket radius results in increased distance between the joint and chain, resulting in more torque produced at the joint. The limiting factors were interference with grasping space due to an overlarge sprocket and the desire to maintain a humanlike form factor; 30 mm was determined to be the maximum allowable phalange thickness. This design approach was applied to all three joint sprockets, though each more distal joint sprocket radius was decreased for the purposes of adaptive grasping. The intraphalangeal sprockets have no effect on chain force output and thus were minimized to meet geometric constraints. The steep redirection angle of these smaller sprockets did not result in high friction due to the ball bearings used, and the 3mm outer diameter of the ball bearings became the limiting factor. 3D printed PLA pulleys were designed as an experimental control for a comparison with a tendon-driven finger. Pulley track radius was matched to the sprocket pitch radius to maintain joint torque forces between transmission mechanisms, and the pulleys were also mounted on bearings.

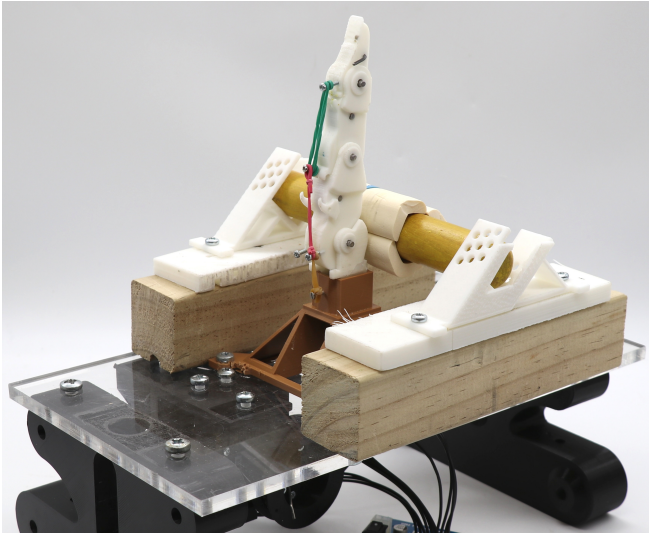


Fig. 5. The test apparatus used for wear cycle testing.

IV. EXPERIMENTAL VALIDATION

A. Experimental Setup

To assess comparative viability and validate the performance of the roller chain transmission system, all experiments were performed on the chain-sprocket design presented as well as an identical control version with the sprockets exchanged for pulleys and the chain for a tendon. Care was taken to ensure geometric parity of critical dimensions, and identical bearings were used. The tendon used in the control finger is a braided Kevlar cord with a rated strength of 91 kg (893 N) [20].

The test rig (Fig. 4) mounted a single finger onto a PLA 3D printed stand such that it pointed vertically in the fully extended position. The stand was mounted to a clear acrylic base sheet with a hole for the finger chain to pass through. The chain was routed through the acrylic base and around a 6 mm radius pulley driven by a Dynamixel XM430-W350-T or MX64AR motor mounted to the underside of the acrylic base sheet. Alternatively, the chain was left free hanging and weights were attached. On the top surface of the acrylic base sheet on both sides two frames were attached, capable of supporting a horizontal rod extending between both stands. This bar was interchangeable with a force dynamometer and in the path of the finger flexion (see Fig. 5).

B. Cycle Testing

Repeated cycles of flexion and extension were performed to assess the longevity and long-term durability of the chain-sprocket transmission system in comparison to the tendon-pulley system. Each finger was in turn secured to the testing rig (Fig. 4) and repeatedly cycled from fully open to closed around a padded wooden dowel. Elastic passive extension elements were replaced as needed. The system was actuated by Dynamixel motors using a Python script and the Dynamixel SDK to control motor position. Disassembly and visual inspection of both transmission systems after cycle



Fig. 6. Tendon wear after 250,000 flexion and extension cycles compared to a new tendon of the same material. Subfig. a) shows the worn tendon and subfig. b) shows a new tendon.

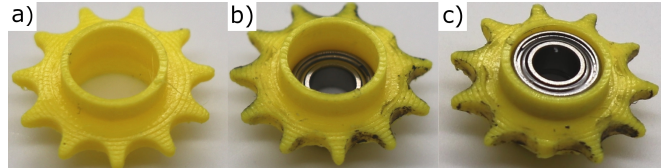


Fig. 7. Comparison of new and used sprockets after 500,000 flexion extension cycles. Subfig. a) shows an unused sprocket, subfig. b) shows the left side of the worn sprocket, and subfig. c) shows the right side of the worn sprocket.

testing revealed significant visual fraying of the tendon after 250,000 cycles of longevity testing. Inspection of the chain at 250,000 cycles showed no sign of wear. The chain finger was cycled further to a total of 500,000 times. Disassembly at this point revealed a broken tooth on the routing sprocket mounted to the proximal phalanx (see Fig. 3). Otherwise, sprockets exhibited no measurable wear but showed some dirt buildup, as shown in Fig. 7. The chain itself exhibited no wear throughout testing.

C. Pinch Grasp Testing

Pinch grip testing was used to compare efficiency, output, and characteristic force responses between the chain-sprocket and tendon-pulley transmission systems. Each of the fingers was placed in the test rig (see Fig. 5) and closed around a force transducer positioned at 45 degrees. This position was chosen as it produced the best output forces out of several tested. Measurements were recorded using the BioPac software, and graphed using Matlab. Figure 9 shows the various pinch grasp force outputs and force response curves of each finger. Notably, underactuated finger designs characteristically underperform on single degree of actuation pinch tests, but the results are still useful for the purposes of comparison. Input force was generated through two methods, a Dynamixel MX-64 servo motor and by hanging weights below the finger. Motor input force was calculated using the measured input current and motor specifications, though anomalies in the recorded data indicated potential inaccuracies in motor output, such as at the jump at 237 N input in Fig. 9. To validate results, further testing was performed on an identical setup by hanging weights from the tendon or chain being tested, though this approach was limited to lower maximum input forces. The pinch grip efficiency of each finger was calculated over the range of input forces

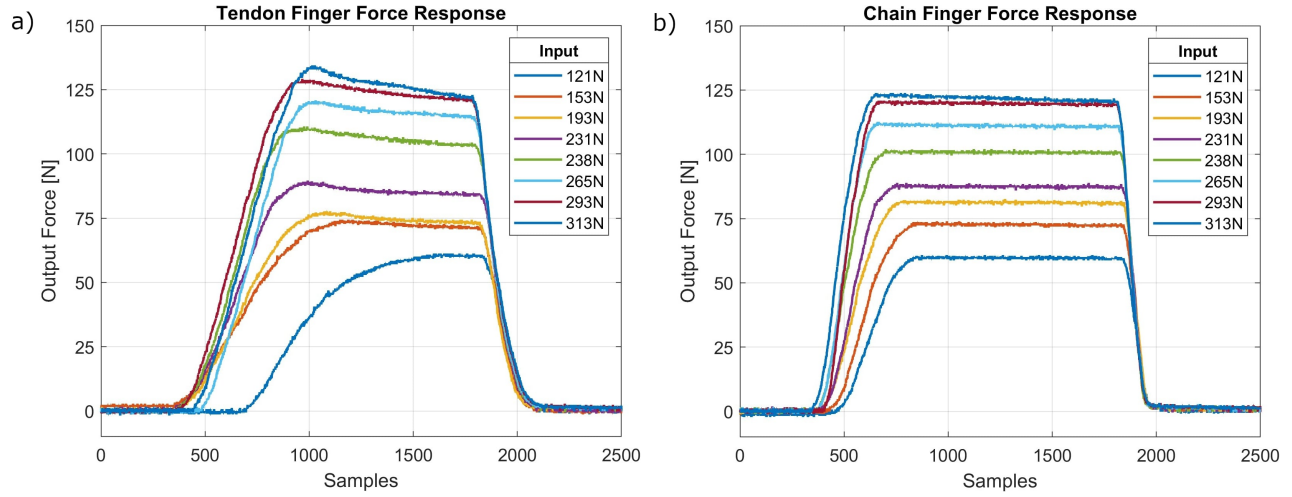


Fig. 8. Measured pinch force response curves of the tendon-driven and chain-driven fingers, plotted across various motor force inputs. Subfigure a) depicts tendon finger response, subfigure b) depicts chain finger response. For each test, the motor was actuated at a constant current for 2 seconds. Input force was calculated from motor specifications and appropriate geometry.

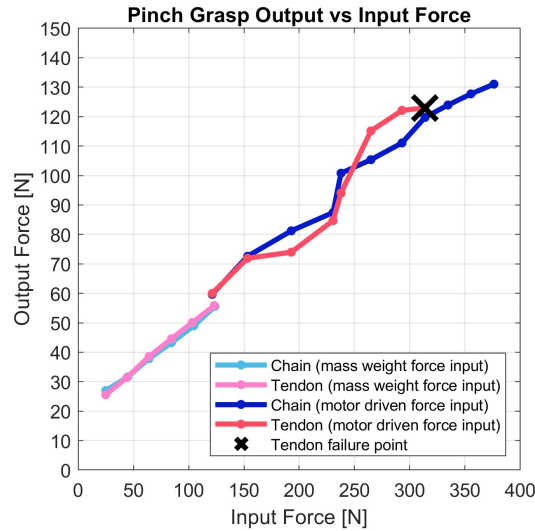


Fig. 9. A comparison of the chain-driven and tendon-driven static pinch grasp force outputs, actuated both by motor and by hanging weights.

from the hanging weight tests; the chain finger had a mean efficiency of 28.0% with a standard deviation of 0.9% while the tendon finger had a mean efficiency of 28.7% with a standard deviation of 2.4%. A two sample t-test results in a P-value of 0.379, indicating a lack of significant statistical difference between the pinch grasp efficiencies of the two transmission methods. Pinch force testing also demonstrated significant differences in maximum force transmission before failure. After 250,000 wear cycles, a tendon was able to withstand 95 N of input force before breaking, a new tendon was able to withstand 334 N before breaking, and the chain after 500,000 wear cycles was able to withstand 418 N before reaching the limits of our test system. The chain finger test produced a maximum pinch grasp output of 131 N before reaching motor maximum and the tendon finger 123 N before failure. It was also found that the tendon exhibited significant

elongation during pinch force testing. Prior to failure, the distance between anchor points during testing increased by 17 mm, from an initial length of 245 mm. Further variation is demonstrated in the differing force response shapes in Fig. 8. As shown, the chain-driven finger exhibits a stable and consistent loading pattern. Conversely, the tendon exhibits a settling period before entering static loading, indicating higher internal elasticity. The rising edges of the tendon force response also exhibit more variation in starting time indicating a small, inconsistent amount of internal slack.

V. CONCLUSIONS AND FUTURE WORK

In this paper we presented a chain-driven design for an adaptive, underactuated prosthetic finger so as to improve durability of prostheses while remaining humanlike and affordable. The proposed design was tested and evaluated against a tendon-driven design of the same dimensions. Input and output forces were measured and wear of components was measured after a high number of flexion/extension cycles. The chain-driven finger was shown to produce a high output force of 131 N without failure, limited only by the input motor. This was superior to the comparable tendon design tested, which failed at a maximum output force of 121 N. In comparison to conventional humanlike tendon design fingers as described in the related works section, the achieved maximum output force was 5-10 times higher. Durability was shown to be comparable or better than tendon designs. Pinch force testing demonstrated further advantages of a chain-driven transmission, displaying no elongation under loading. The results presented indicate a high degree of potential for further designs incorporating chain-driven finger transmissions.

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